FINAL ANALYSIS REPORT

Formula Definition

$$L = L_{\text{linear}} \left[e^{\left(\frac{345}{509} \frac{U\,T}{L_{\text{linear}}}\right)} + \frac{1052\,\pi\,U}{1052\,\pi\,U + 435\,g\,T} - \frac{5\tanh\left(\frac{U\,T}{L_{\text{linear}}} \ln\left(T\sqrt{\frac{g}{d}}\right) + \frac{2117}{961}\right)}{67\ln\left(\frac{H}{d}\right)} \right]$$

$$L_{\text{linear}} = \frac{g\,T^2}{2\,\pi} \tanh\left(\frac{2\,\pi\,d}{L_{\text{linear}}}\right)$$

$$C_0 = \frac{g\,T}{2\,\pi} \approx 1.5608\,T$$

$g \approx 9.80665\ \text{m/s}^2$ and ln denotes the natural logarithm.

Global Accuracy

Metric	Value
MAPE (Mean % Error)	2.3798%
Max Error Percent	80.1832%
RMSE (Average Unit Error)	2.890977
Bias (Mean Residual)	0.518625

Percentile Breakdown

Coverage Level	Absolute Percentage Error
50% of data	< 2.0505%
90% of data	< 4.7842%
99% of data	< 7.2175%

Regime Analysis

Regime	Count	MAPE	Max Error
Shallow	3341	2.3153%	8.7899%
Intermediate	2793	2.4077%	80.1832%
Deep	3482	2.4194%	30.0519%

Top 20 Worst Predictions

Index	H	T	d	U	L True	L Pred	Diff	Error %
1311	2.84	4.72	14.10	-2.35	38.07	7.54	30.53	80.18
38	1.04	4.90	49.87	-1.91	12.40	16.12	-3.73	30.05
609	1.07	6.71	35.78	-2.49	28.01	32.75	-4.74	16.94

Index	H	T	d	U	L True	L Pred	Diff	Error %
272	1.08	6.08	17.45	-2.35	20.93	24.31	-3.38	16.16
459	2.68	6.38	32.98	-2.42	25.19	29.14	-3.95	15.69
249	2.21	5.75	45.28	-2.19	20.21	23.36	-3.15	15.57
416	1.00	6.15	29.42	-2.27	24.07	27.76	-3.69	15.33
298	2.17	5.85	25.02	-2.20	21.54	24.79	-3.25	15.08
68	1.98	4.74	17.61	-1.78	14.32	16.47	-2.15	15.03
27	1.04	4.28	43.77	-1.58	11.73	13.45	-1.73	14.75
230	2.36	5.62	37.97	-2.14	19.74	22.42	-2.68	13.58
415	2.55	6.03	33.38	-2.23	24.07	27.08	-3.01	12.50
44	1.40	4.49	47.27	-1.69	12.85	14.41	-1.57	12.20
245	1.04	5.52	27.23	-2.04	20.05	22.42	-2.37	11.83
821	1.44	6.77	33.25	-2.43	31.40	35.03	-3.63	11.56
968	1.05	6.81	38.09	-2.35	33.45	37.22	-3.76	11.25
738	2.27	6.79	7.63	-2.31	30.23	26.85	3.38	11.19
201	1.51	5.56	5.05	-1.97	19.04	16.93	2.11	11.07
3181	4.54	5.56	10.59	2.21	60.49	67.15	-6.66	11.01
4368	4.58	9.33	12.75	-2.22	74.28	66.18	8.10	10.90